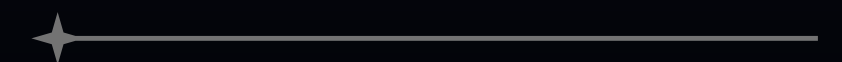


CLIMATE CHANGE TREND ANALYSIS AND FORECASTING



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PROJECT OBJECTIVE AND DATASET OVERVIEW

The main objective of this project is to :

- Analyze climate change trends using historical greenhouse gas emission data.
- Identify emission patterns across different countries.
- Develop forecasting models for future emissions.

Dataset Overview :

- OWID CO₂ Dataset

Features :

- Country
- Year
- CO₂ emissions
- CO₂ per capita
- Methane
- Nitrous oxide
- Total greenhouse gas emissions (ghg)

notebook >	owid-co2-data.csv
1	country,year,iso_code,population,gdp,cement_co2,cement_co2_per_capita,co2,co2_growth_abs,co2_growth_prct,co2_inc
2	Afghanistan,1750,AFG,2802560,,0.000,0.000,,,,,,,,,0.000,,,,,,,,,
3	Afghanistan,1751,AFG,,,0.000,,,,,,,,,0.000,,,,,,,,,
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20	Afghanistan,1768,AFG,,,0.000,,,,,,,,,0.000,,,,,,,,,

Data Acquisition, Exploration & EDA

Data Preparation :

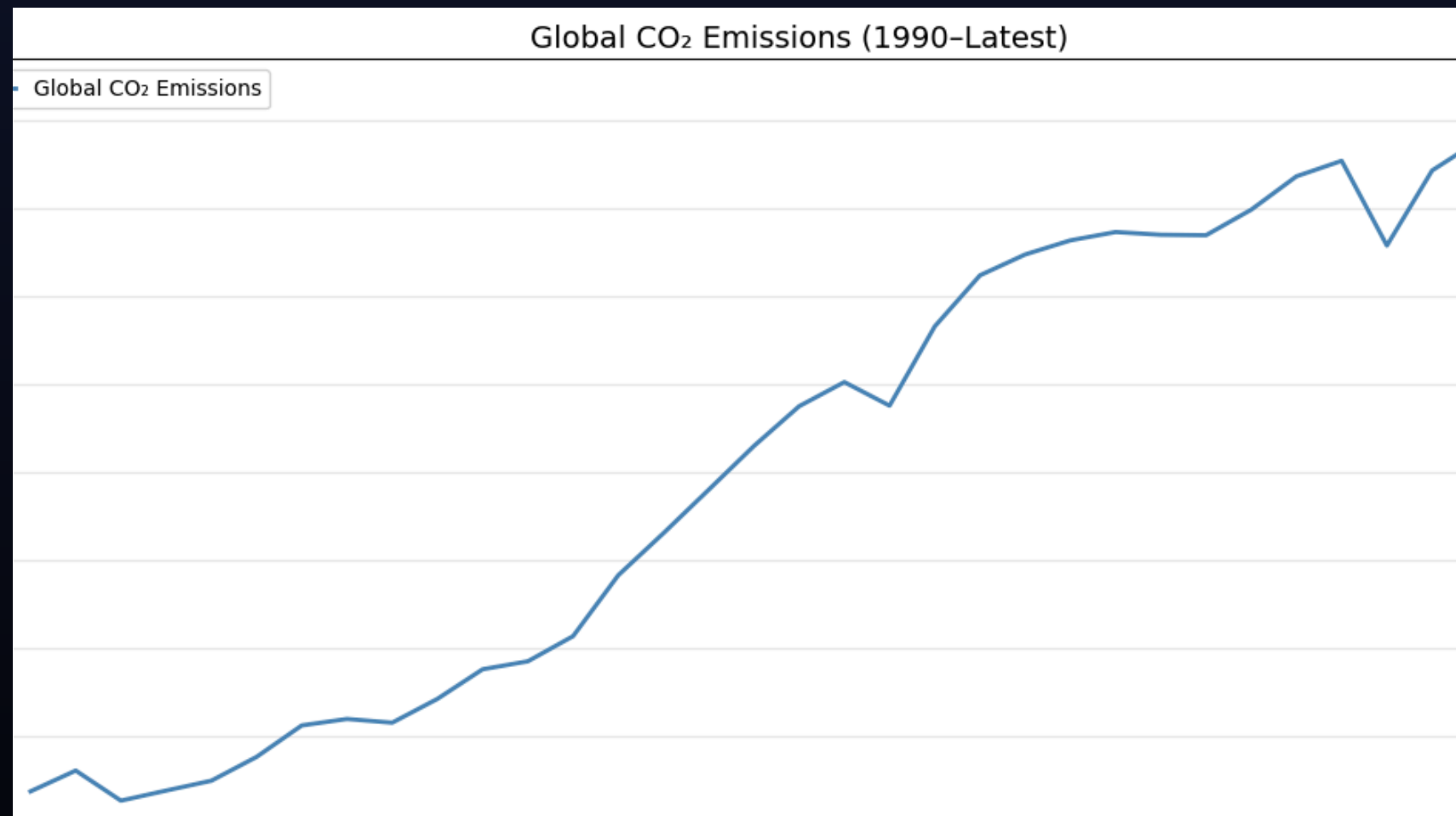
- Examined dataset structure: rows, columns, data types, and key variables.
- Analyzed missing values and data coverage.
- Data filtering was done year ≥ 1990 and removed aggregate regions (e.g. Asia, world) and sovereign countries were selected. EDA was performed.

Key Findings :

- Global CO₂ emissions show an overall increasing trend.
 - China has the highest emission levels, while India shows rapid growth.
 - CO₂ remains the dominant contributor to total greenhouse gas emissions.
-

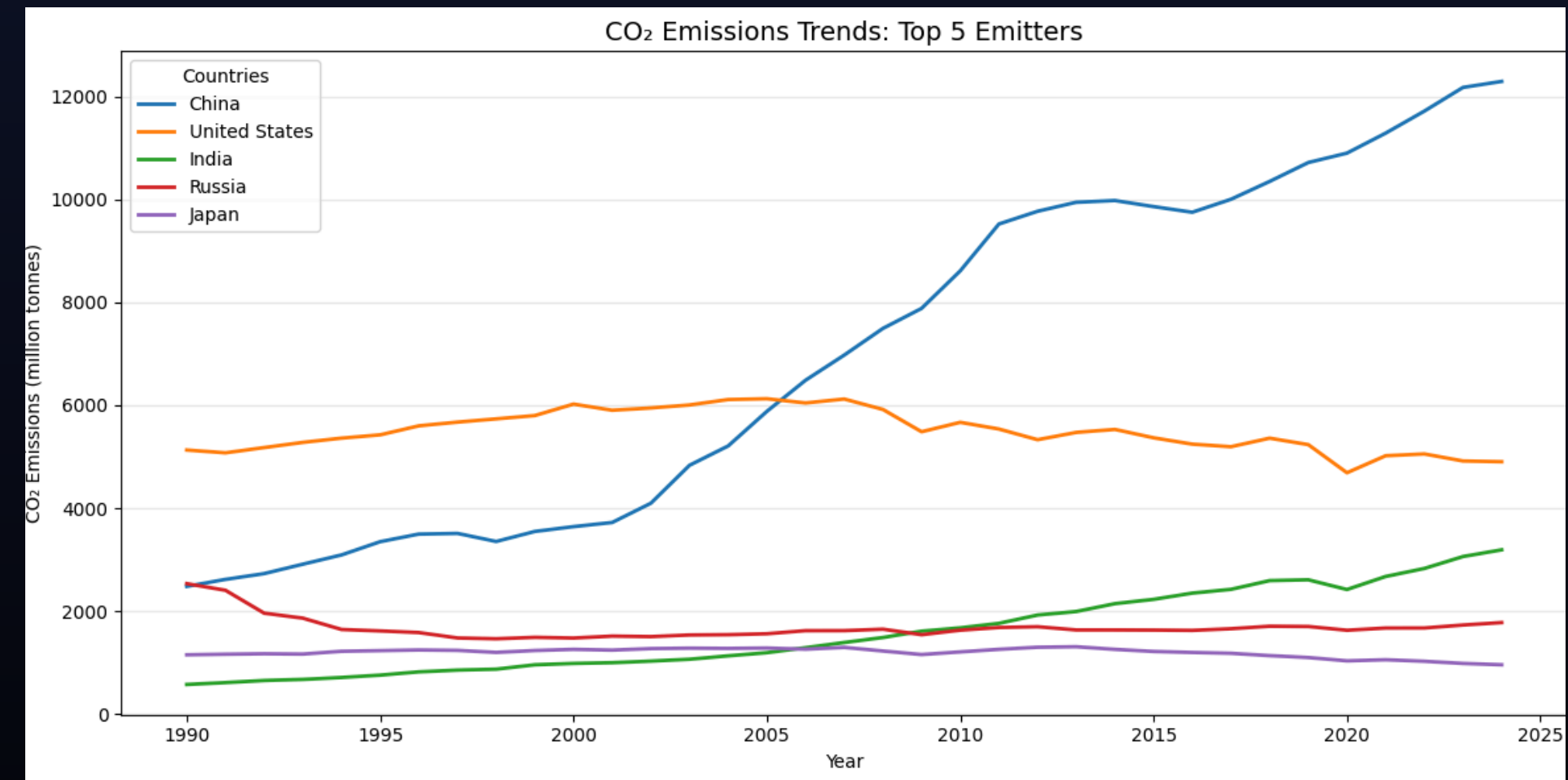
EDA KEY FINDINGS

1. Global CO₂ Emission Trend



- Global CO₂ emissions show an overall increasing trend over the study period.
- The rise reflects increasing industrial activity, population growth, and energy consumption

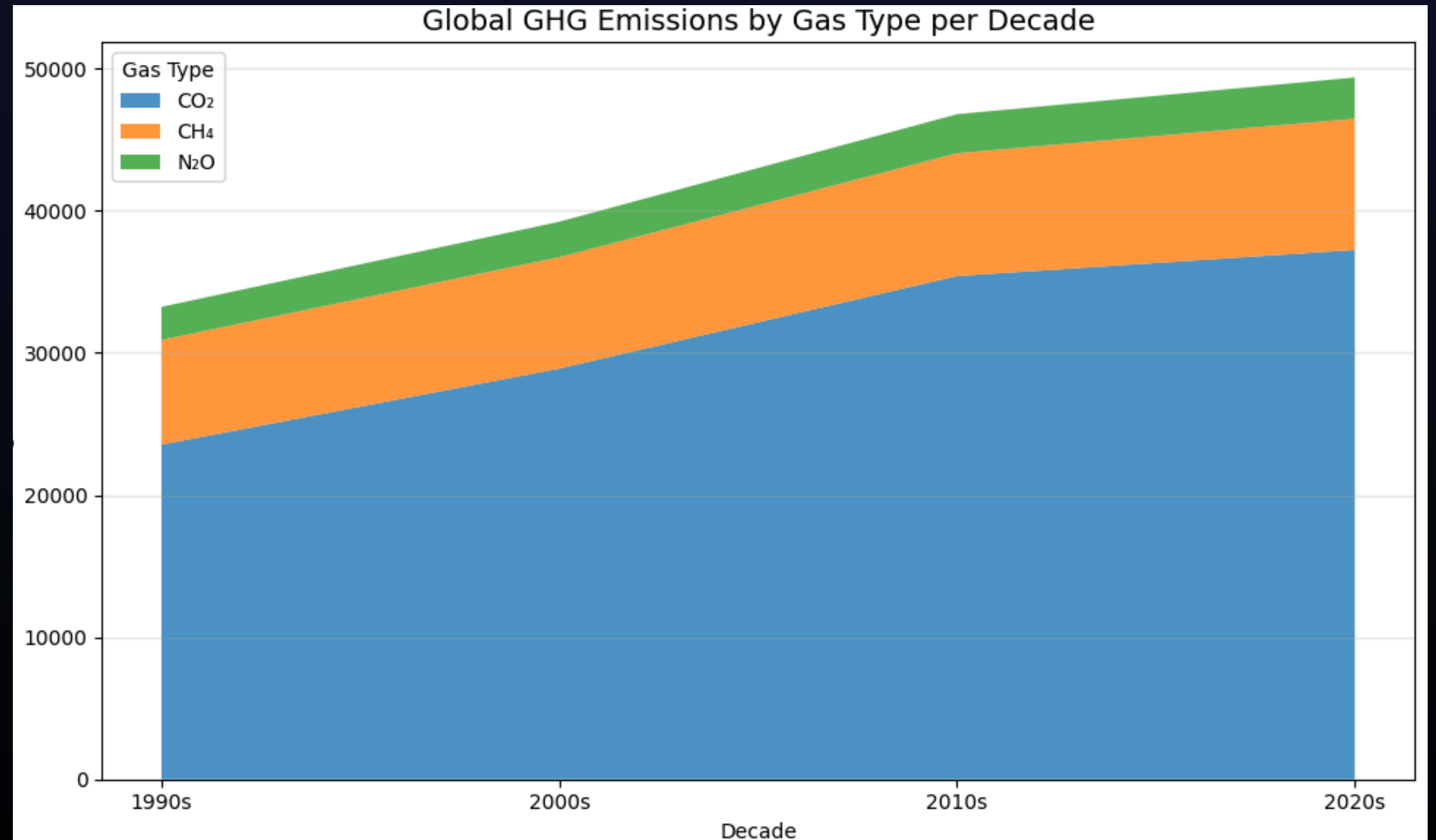
2. Top CO₂ Emitting Countries



- The USA remains one of the major contributors to global CO₂ emissions.
- Emissions vary significantly across countries based on population, industrial activity, and energy consumption.

3. Greenhouse Gas Composition

- CO₂ is the dominant contributor among greenhouse gases.
- Methane and nitrous oxide contribute smaller but important shares.
- Different gases show different emission patterns over time.



FEATURE ENGINEERING

Time-Based Features

years_since_1990

- Created to represent the number of years from the starting point (1990).
- Helps forecasting models capture the overall long-term emission trend.

decade

- Categorizes years into decades to analyze changes across different time periods.
- Helps identify broader emission patterns over time.

Rolling Statistics

5-Year Rolling Mean

- Calculated the average emission value over a 5-year window.
- Reduces yearly fluctuations and highlights the underlying trend.

Lag Features

- Created lag variables: co2_lag1, co2_lag2, and co2_lag3 to represent previous years' CO₂ emissions.
- These features capture historical emission patterns and help forecasting models learn from past trends.
- Lag features are useful in time-series prediction because previous observations often influence future values.

BASELINE ML MODELS



Naive baseline model



- Predicts future CO₂ emissions using the current year's value.
- Used as a simple reference model for comparison.



Linear regression



Captures linear relationships between time-based features and emissions.



Random forest regressor



Captures nonlinear patterns and feature interactions.

Objective: Predict next year's CO₂ emissions using historical emission patterns and engineered features.

Target variable: CO₂ emissions (co2)

Evaluation :

- Used time-based train-test split:
- Training: 1990–2018
- Testing: 2019–2023

Metrics:

- MAE
- RMSE

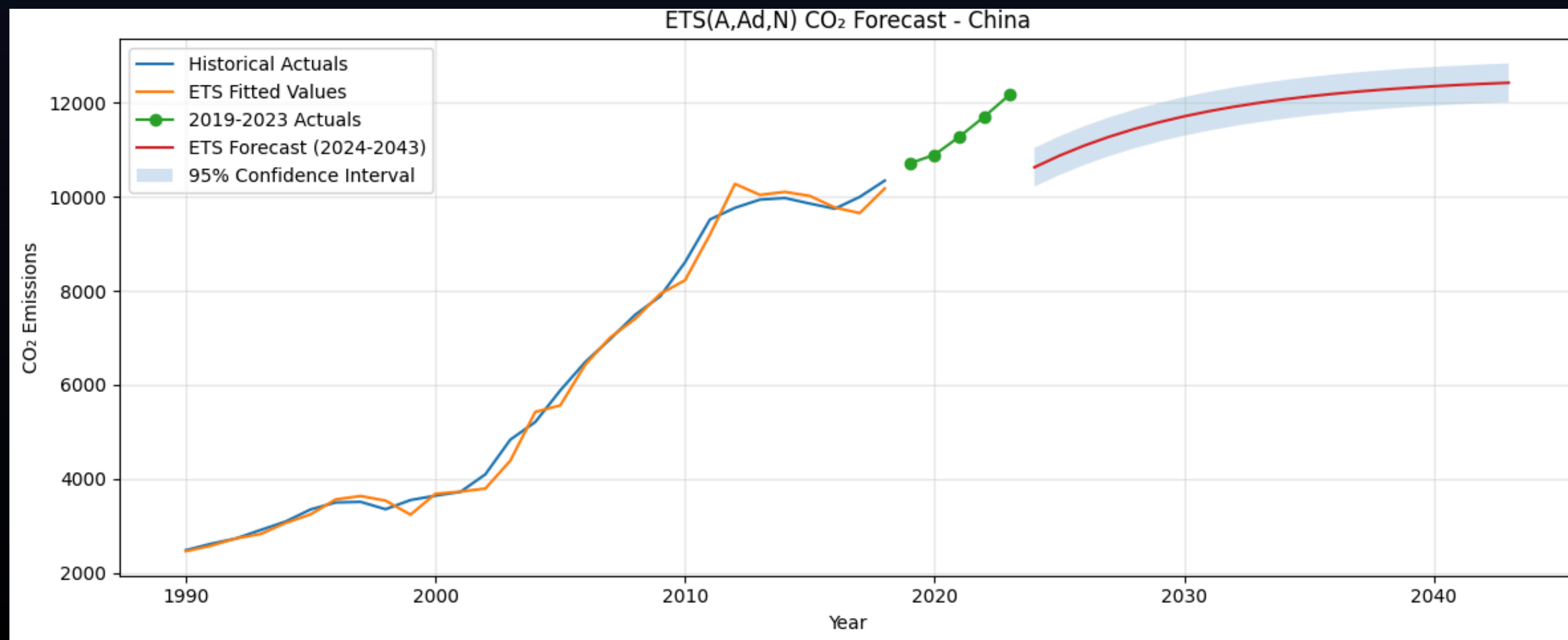
Model comparison table (MAE & RMSE)

	country	Baseline MAE	Baseline RMSE	LR MAE	LR RMSE	RF MAE	RF RMSE
0	China	380.264314	380.264314	122.477459	123.491460	964.943285	1037.822098
1	India	210.551410	210.551410	18.558444	21.881422	156.375047	184.781198
2	Japan	42.735691	42.735691	5.161882	5.666255	103.518975	107.144657
3	United States	326.759549	326.759549	33.666872	42.815094	279.267180	341.888192
4	Russia	51.010591	51.010591	3.071546	3.548230	11.808643	13.733550
5	Germany	50.791896	50.791896	7.732684	9.130234	94.110955	96.820607
6	United Kingdom	26.115424	26.115424	6.943559	8.348277	51.632507	56.147720
7	Brazil	29.459077	29.459077	6.385103	6.977136	9.523640	11.830554
8	Canada	28.667970	28.667970	0.423022	0.536373	15.183607	17.397895
9	Australia	9.862915	9.862915	0.494231	0.713844	11.734588	12.999249

ETS Forecasting Results

ETS(A,Ad,N) Model :

- Implemented Exponential Smoothing with: Additive Error, Additive Damped Trend, & No Seasonality
- It is suitable for annual CO₂ emission data because:
 - No within-year seasonal pattern exists.
 - Damped trend avoids unrealistic long-term growth.
 - Works well with limited historical data.



Forecast Summary Table

	Country	2030 Forecast (MtCO ₂)	2035 Forecast (MtCO ₂)	2040 Forecast (MtCO ₂)	2020 Actual	% Change 2020 → 2040
0	Australia	425.560937	429.373841	431.519888	398.547	8.273275
1	Brazil	513.931816	536.130246	555.198411	447.999	23.928493
2	Canada	580.281976	583.199981	585.165613	524.208	11.628516
3	China	11714.071703	12138.534046	12353.746265	10896.521	13.373308
4	Germany	745.610619	726.166555	709.584467	647.177	9.643029
5	India	3401.084264	3974.680846	4534.080197	2422.732	87.147410
6	Japan	1139.033390	1139.147243	1139.193616	1037.285	9.824553
7	Russia	1713.694794	1720.965387	1723.347815	1631.902	5.603634
8	United Kingdom	274.480687	212.720255	158.107101	326.263	-51.539984
9	United States	5257.378811	5241.980070	5234.324876	4689.954	11.607169

Key Observations :

- ETS effectively captures long-term emission trends.
- Forecast uncertainty increases as the prediction horizon extends.

KEY TAKEAWAYS & LIMITATIONS

Key Takeaways:

- Global CO₂ emissions have shown an overall increasing trend since 1990.
- Linear Regression achieved the best performance among the evaluated models.
- The ETS model provided realistic long-term forecasts by capturing historical emission trends with a damped trend.

Limitations:

- Forecasts are based only on historical emission data and trends.
- Prediction accuracy depends on the quality and completeness of the available dataset.





THANK
YOU!

